

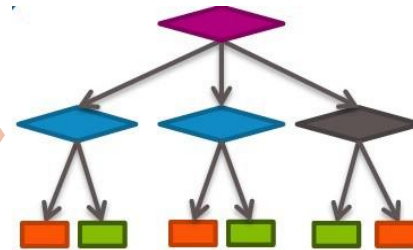
Lecture 4: Naïve Bayes

Decision Tree: Recap

Information of Cars

- Auction
- Color
- MMRCurrentAuctionAveragePrice
- Size
- TopThreeAmericanName
- VehBCost
- VehicleAge
- VehOdo
- WarrantyCost
- WheelType

Predictors (X)



Classifier

IsBadBuy

Target variable
/Label (Y)

Step 1: Kicked Vehicle Data

	Auction	Color	IsBadBuy	MMRCurrentAuctionAveragePrice	Size	TopThreeAmericanName	VehBCost	VehicleAge	VehOdo	WarrantyCost	WheelType
0	ADESA	WHITE	No	2871	LARGE TRUCK	FORD	5300	8	75419	869	Alloy
1	ADESA	GOLD	Yes	1840	VAN	FORD	3600	8	82944	2322	Alloy
2	ADESA	RED	No	8931	SMALL SUV	CHRYSLER	7500	4	57338	588	Alloy
3	ADESA	GOLD	No	8320	CROSSOVER	FORD	8500	5	55909	1169	Alloy
4	ADESA	GREY	No	11520	LARGE TRUCK	FORD	10100	5	86702	853	Alloy
...	...	...	...	...	...	...	...	...	...	...	...
9995	ADESA	RED	No	7536	SMALL SUV	CHRYSLER	6600	4	85377	983	Alloy
9996	ADESA	BLACK	No	4921	LARGE TRUCK	GM	7000	7	89665	1543	Alloy
9997	ADESA	BLACK	No	9263	MEDIUM SUV	CHRYSLER	9000	4	59383	1417	Alloy
9998	ADESA	BLUE	No	3240	MEDIUM	OTHER	5500	4	48642	482	Covers
9999	ADESA	SILVER	No	6148	MEDIUM SUV	FORD	7000	5	82563	1243	Alloy

10000 rows x 11 columns

Step 2: Data Preprocessing

Split the data to training and testing sets

	IsBadBuy	MMRCurrentAuctionAveragePrice	VehBCost	VehicleAge	VehOdo	WarrantyCost	Auction_MANHEIM	Auction_OTHER	Color_BLACK	...	WheelType_unkwnWheel	
0	No	2871	5300	8	75419	869	0	0	0		0	
1	Yes	1840	3600	8	82944	2322	0	0	0		0	
2	No	8931	7500	4	57338	588	0	0	0		0	
3	No	8320	8500	5	55909	1169	0	0	0		0	
4	No	11520	10100	5	86702	853	0	0	0		0	
...	...	...	...	...	...	...	...	...	...	...	...	
6995	No	2762	4185	4	84379	1243	0	1	0		0	
6996	Yes	5144	8705	5	80600	1373	0	1	0		0	
6997	No	2912	4485	6	77686	1389	0	1	0		0	
6998	No	4965	6495	4	66731	1020	0	1	0		0	
6999	No	2724	3985	6	86275	1243	0	1	0		0	
...	...	...	...	...	...	...	...	...	...	...	...	
9995	No	7536	6600	4	85377	983	0	0	0		0	
9996	No	4921	7000	7	89665	1543	0	0	1		0	
9997	No	9263	9000	4	59383	1417	0	0	1		0	
9998	No	3240	5500	4	48642	482	0	0	0		0	
9999	No	6148	7000	5	82563	1243	0	0	0		0	

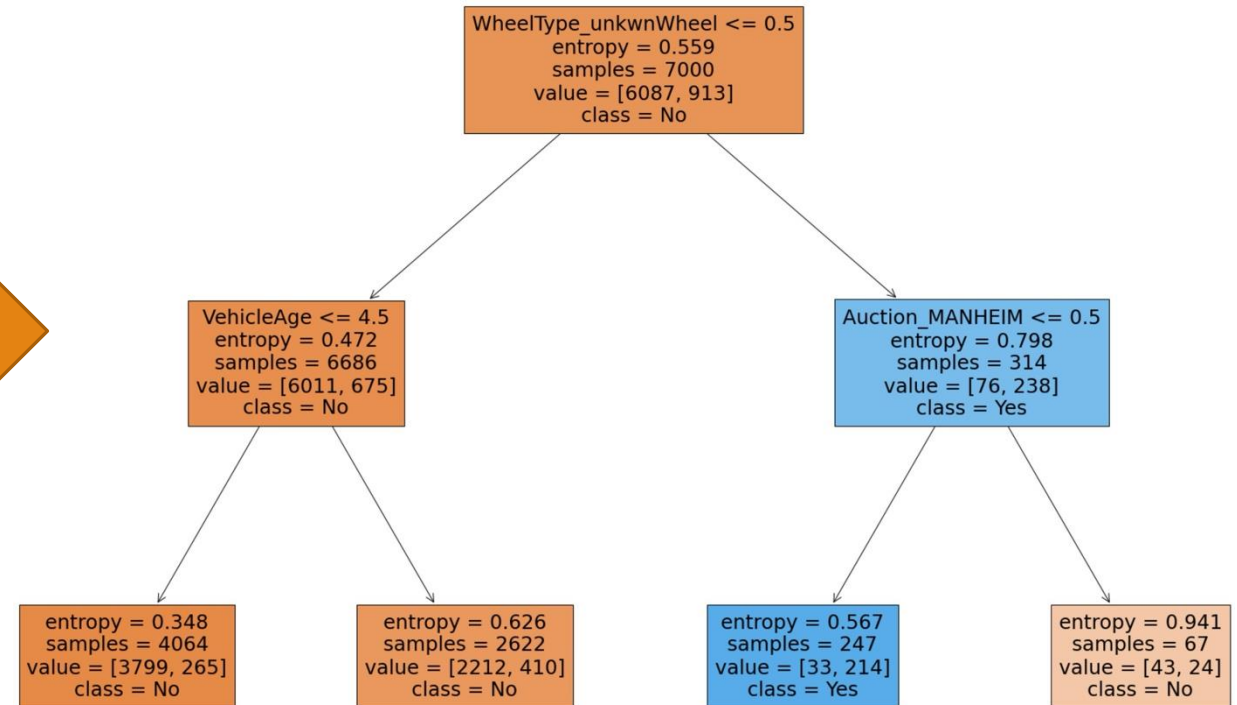
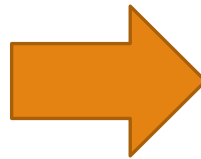
Training data

Testing data

Step 3: Build a decision tree on training data

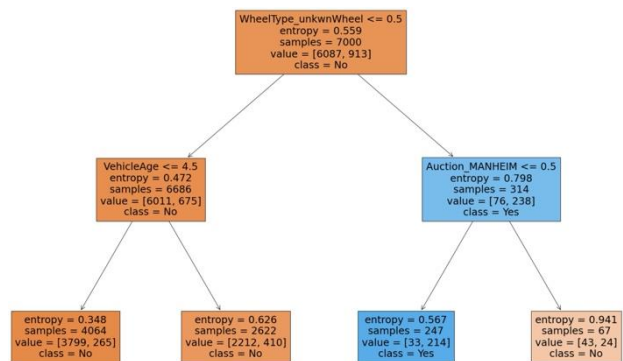
	IsBadBuy	MMRCurrentAuctionAveragePrice	VehBCost	VehicleAge	...
0	No	2871	5300	8	
1	Yes	1840	3600	8	
2	No	8931	7500	4	
3	No	8320	8500	5	
4	No	11520	10100	5	
...	...	...	...	...	...
6995	No	2762	4185	4	
6996	Yes	5144	8705	5	
6997	No	2912	4485	6	
6998	No	4965	6495	4	
6999	No	2724	3985	6	

Training data



Decision Tree Model

Step 4: Evaluate Decision Tree Model Performance



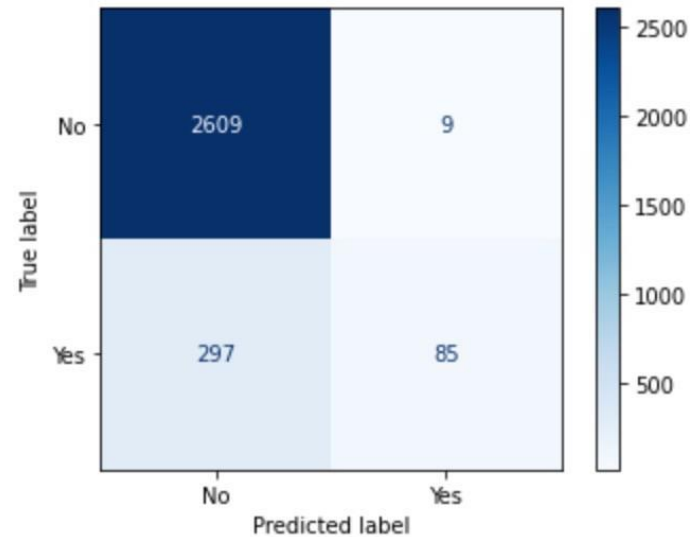
...	...	...	...	...	...	...	...	...	...
9995		7536	6600	4	85377	983	0	0	0
9996		4921	7000	7	89665	1543	0	0	1
9997		9263	9000	4	59383	1417	0	0	1
9998		3240	5500	4	48642	482	0	0	0
9999		6148	7000	5	82563	1243	0	0	0

Testing data

Predictions	Real value
IsBadBuy	IsBadBuy
No	No
No	No
No	No
No	No
Yes	No
Yes	No

Step 4: Evaluate Decision Tree Model Performance

Performance on the **testing data**:



	precision	recall	f1-score	support	
No	0.90	1.00	0.94	2618	Overall performance on Not bad buy class
Yes	0.90	0.22	0.36	382	
accuracy			0.90	3000	Overall performance on bad buy class
macro avg	0.90	0.61	0.65	3000	
weighted avg	0.90	0.90	0.87	3000	
					Model's overall performance

Naïve Bayes

Overview

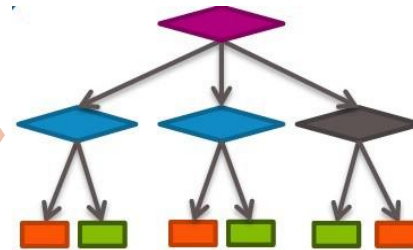
- Naïve Bayes Classifier
- Model Comparison: Decision Tree and Naïve Bayes Model
- Cost-benefit Analysis

Naïve Bayes

Information of Cars

- Auction
- Color
- MMRCurrentAuctionAveragePrice
- Size
- TopThreeAmericanName
- VehBCost
- VehicleAge
- VehOdo
- WarrantyCost
- WheelType

Predictors (X)



Classifier

Decision Tree or Naïve Bayes

IsBadBuy

Target variable
/Label (Y)

Naïve Bayes

- Model a classification rule directly
 - Decision Tree: rule-based model
- Make a probabilistic model of data within each class
 - Naïve Bayes: probabilistic model

Naïve Bayes

- The **Naive Bayes** algorithm describes a simple method to apply Bayes' theorem to classification problems.
 - The Naive Bayes algorithm is named as such because it makes some "naive" assumptions about the data.
 - Naive Bayes assumes that all of the variables in the dataset are equally important and independent. (strong assumption)
 - However, in most cases when these assumptions are violated, Naive Bayes still performs fairly well.
- Naive assumptions + Bayes' theorem

Naïve Bayes

- Bayes' theorem, named after 18th-century British mathematician Thomas Bayes, is a mathematical formula for determining conditional probability.

$$P(C|A) = \frac{P(A, C)}{P(A)} = \frac{P(A|C)P(C)}{P(A)}$$

Diagram illustrating Bayes' theorem formula with labels:

- $P(C|A)$: posterior
- $P(A, C)$: likelihood
- $P(A)$: marginal probability
- $P(A|C)$: prior
- $P(C)$: prior

Naïve Bayes

- You are planning a picnic today, but the morning is cloudy
 - Cloudy mornings are common (about 40% of days start cloudy)
 - And this is usually a dry month (only 3 of 30 days tend to be rainy, or 10%)
 - Rainy days must start off cloudy
- $P(\text{Rain} | \text{Cloudy}) = \frac{P(\text{Cloudy} | \text{Rain}) P(\text{Rain})}{P(\text{Cloudy})}$

Naïve Bayes

- You are planning a picnic today, but the morning is cloudy
 - Cloudy mornings are common (about 40% of days start cloudy)
 - And this is usually a dry month (only 3 of 30 days tend to be rainy, or 10%)
 - Rainy days must start off cloudy

- $$P(\text{Rain} | \text{Cloudy}) = \frac{P(\text{Cloudy} | \text{Rain}) P(\text{Rain})}{P(\text{Cloudy})} = \frac{1 * 0.1}{0.4} = 0.25$$

Kicked Vehicle

- Figure out which cars have a higher risk of being kick can provide real value to dealerships trying to provide the best inventory selection possible to their customers.
- Problem: Predict if the car purchased at the Auction is a Kick (bad buy).

Kicked Vehicle

	Auction	Color	IsBadBuy	MMRCurrentAuctionAveragePrice	Size	TopThreeAmericanName	VehBCost	VehicleAge	VehOdo	WarrantyCost	WheelType
0	ADESA	WHITE	No	2871	LARGE TRUCK	FORD	5300	8	75419	869	Alloy
1	ADESA	GOLD	Yes	1840	VAN	FORD	3600	8	82944	2322	Alloy
2	ADESA	RED	No	8931	SMALL SUV	CHRYSLER	7500	4	57338	588	Alloy
3	ADESA	GOLD	No	8320	CROSSOVER	FORD	8500	5	55909	1169	Alloy
4	ADESA	GREY	No	11520	LARGE TRUCK	FORD	10100	5	86702	853	Alloy
...	...	...	...	...	...	...	...	...	...	...	...
9995	ADESA	RED	No	7536	SMALL SUV	CHRYSLER	6600	4	85377	983	Alloy
9996	ADESA	BLACK	No	4921	LARGE TRUCK	GM	7000	7	89665	1543	Alloy
9997	ADESA	BLACK	No	9263	MEDIUM SUV	CHRYSLER	9000	4	59383	1417	Alloy
9998	ADESA	BLUE	No	3240	MEDIUM	OTHER	5500	4	48642	482	Covers
9999	ADESA	SILVER	No	6148	MEDIUM SUV	FORD	7000	5	82563	1243	Alloy

10000 rows x 11 columns

P(IsBadBuy=Yes | WheelType=unkwnWheel, Auction=OTHER, Color = GOLD, Size = VAN, TopThreeAmericanName = FORD, VehBCost = 3600, VehicleAge = 8, VehOdo = 24244, WarrantyCost = 1400)

P(IsBadBuy=No | WheelType=unkwnWheel, Auction=OTHER, Color = GOLD, Size = VAN, TopThreeAmericanName = FORD, VehBCost = 3600, VehicleAge = 8, VehOdo = 24244, WarrantyCost = 1400)

Naïve Bayes

- WheelType=unkwnWheel, Auction=OTHER
- IsBadBuy = Yes or No?

■ $P(\text{IsBadBuy=Yes} \mid \text{WheelType=unkwnWheel}, \text{Auction=OTHER})$

◦ $= \frac{P(\text{WheelType=unkwnWheel}, \text{Auction=OTHER} \mid \text{IsBadBuy=Yes}) * P(\text{IsBadBuy=Yes})}{P(\text{WheelType=unkwnWheel}, \text{Auction=OTHER})}$

Bayes' theorem

◦ $\propto P(\text{WheelType} = \text{unkwnWheel}, \text{Auction} = \text{OTHER} \mid \text{IsBadBuy} = \text{Yes}) * P(\text{IsBadBuy} = \text{Yes})$

◦ $= P(\text{WheelType=unkwnWheel} \mid \text{IsBadBuy=Yes}) * P(\text{Auction=OTHER} \mid \text{IsBadBuy=Yes}) * P(\text{IsBadBuy=Yes})$

Conditional independence assumption

■ $P(\text{IsBadBuy=No} \mid \text{WheelType=unkwnWheel}, \text{Auction=OTHER})$

◦ $= \frac{P(\text{WheelType=unkwnWheel}, \text{Auction=OTHER} \mid \text{IsBadBuy=No}) * P(\text{IsBadBuy=No})}{P(\text{WheelType=unkwnWheel}, \text{Auction=OTHER})}$

◦ $\propto P(\text{WheelType} = \text{unkwnWheel}, \text{Auction} = \text{OTHER} \mid \text{IsBadBuy} = \text{No}) * P(\text{IsBadBuy} = \text{No})$

◦ $= P(\text{WheelType=unkwnWheel} \mid \text{IsBadBuy=No}) * P(\text{Auction=OTHER} \mid \text{IsBadBuy=No}) * P(\text{IsBadBuy=No})$

Naïve Bayes

WheelType	Auction	IsBadBuy
Alloy	OTHER	Yes
Special	ADESA	No
Alloy	MANHEIM	No
unkwnWheel	OTHER	No
unkwnWheel	OTHER	Yes

■ IsBadBuy = Yes (40%; 2 instances)

WheelType:	Alloy	1
	Special	0
	unkwnWheel	1
Auction:	ADESA	0
	MANHEIM	0
	OTHER	2

Prediction for (WheelType=unkwnWheel, Auction=OTHER)

■ $P(\text{IsBadBuy} = \text{Yes} \mid \text{WheelType}=\text{unkwnWheel}, \text{Auction}=\text{OTHER})$

$$\propto P(\text{WheelType}=\text{unkwnWheel} \mid \text{IsBadBuy} = \text{Yes}) * P(\text{Auction}=\text{OTHER} \mid \text{IsBadBuy} = \text{Yes}) * P(\text{IsBadBuy} = \text{Yes})$$

$$= 0.5 * 1 * 0.4 = 0.2$$

■ $P(\text{IsBadBuy} = \text{No} \mid \text{WheelType}=\text{unkwnWheel}, \text{Auction}=\text{OTHER})$

$$\propto P(\text{WheelType}=\text{unkwnWheel} \mid \text{IsBadBuy} = \text{No}) * P(\text{Auction}=\text{OTHER} \mid \text{IsBadBuy} = \text{No}) * P(\text{IsBadBuy} = \text{No})$$

$$= 0.333 * 0.333 * 0.6 = 0.0665$$

■ IsBadBuy = No (60%; 3 instances)

WheelType:	Alloy	1
	Special	1
	unkwnWheel	1
Auction:	ADESA	1
	MANHEIM	1
	OTHER	1

Naïve Bayes

- Predicted probabilities:

- $P(\text{IsBadBuy} = \text{Yes} \mid \text{WheelType}=\text{unkwnWheel}, \text{Auction}=\text{OTHER}) = \frac{0.2}{0.2665}$
- $P(\text{IsBadBuy} = \text{No} \mid \text{WheelType}=\text{unkwnWheel}, \text{Auction}=\text{OTHER}) = \frac{0.0665}{0.2665}$

Naïve Bayes

WheelType	Auction	IsBadBuy
Alloy	OTHER	Yes
Special	ADESA	No
Alloy	MANHEIM	No
unkwnWheel	OTHER	No
unkwnWheel	OTHER	Yes

Prediction for (WheelType=Special, Auction=OTHER)

- zero value causes the posterior to be zero
- $P(\text{IsBadBuy} = \text{Yes} \mid \text{WheelType}=\text{Special}, \text{Auction}=\text{OTHER})$
 $\propto P(\text{WheelType}=\text{Special} \mid \text{IsBadBuy} = \text{Yes}) * P(\text{Auction}=\text{OTHER} \mid \text{IsBadBuy} = \text{Yes}) * P(\text{IsBadBuy} = \text{Yes})$
 $= 0 * 1 * 0.4 = 0$
 - $P(\text{IsBadBuy} = \text{No} \mid \text{WheelType}=\text{Special}, \text{Auction}=\text{OTHER})$
 $\propto P(\text{WheelType}=\text{Special} \mid \text{IsBadBuy} = \text{No}) * P(\text{Auction}=\text{OTHER} \mid \text{IsBadBuy} = \text{No}) * P(\text{IsBadBuy} = \text{No})$
 $= 0.333 * 0.333 * 0.6 = 0.0665$

■ IsBadBuy = Yes (40%; 2 instances)

WheelType:	Alloy	1
	Special	0
	unkwnWheel	1
Auction:	ADESA	0
	MANHEIM	0
	OTHER	2

■ IsBadBuy = No (60%; 3 instances)

WheelType:	Alloy	1
	Special	1
	unkwnWheel	1
Auction:	ADESA	1
	MANHEIM	1
	OTHER	1

Naïve Bayes

- **Laplace estimator/Laplace smoothing**
 - The Laplace estimator essentially adds a small number to each of the counts, which ensures that each feature has a nonzero probability of occurring with each class.
 - Typically, the Laplace estimator is set to 1, which ensures that each class-feature combination is found in the data at least once.
 - In practice, given a large enough training dataset, this Laplace estimator is unnecessary and the value of 1 is almost always used.
 - Laplace smoothing is useful especially when the dataset is small

Naïve Bayes

WheelType	Auction	IsBadBuy
Alloy	OTHER	Yes
Special	ADESA	No
Alloy	MANHEIM	No
unkwnWheel	OTHER	No
unkwnWheel	OTHER	Yes

Prediction for (WheelType=Special, Auction=OTHER)

- $P(\text{IsBadBuy} = \text{Yes} \mid \text{WheelType}=\text{Special}, \text{Auction}=\text{OTHER})$**
 $\propto P(\text{WheelType}=\text{Special} \mid \text{IsBadBuy} = \text{Yes}) * P(\text{Auction}=\text{OTHER} \mid \text{IsBadBuy} = \text{Yes}) * P(\text{IsBadBuy} = \text{Yes})$
 $= 0.2 * 0.6 * 0.4 = 0.048$
- $P(\text{IsBadBuy} = \text{No} \mid \text{WheelType}=\text{Special}, \text{Auction}=\text{OTHER})$**
 $\propto P(\text{WheelType}=\text{Special} \mid \text{IsBadBuy} = \text{No}) * P(\text{Auction}=\text{OTHER} \mid \text{IsBadBuy} = \text{No}) * P(\text{IsBadBuy} = \text{No})$
 $= 0.333 * 0.333 * 0.6 = 0.0665$

■ IsBadBuy = Yes (40%; 2 instances)

WheelType:	Alloy	2
	Special	1
	unkwnWheel	2
Auction:	ADESA	1
	MANHEIM	1
	OTHER	3

■ IsBadBuy = No (60%; 3 instances)

WheelType:	Alloy	2
	Special	2
	unkwnWheel	2
Auction:	ADESA	2
	MANHEIM	2
	OTHER	2

Naïve Bayes

- Prior Probability: $P(C)$
 - C is the target variable (categorical variable)
 - $P(C)$ is easy to calculate
 - $P(\text{IsBadBuy} = \text{Yes}) = 0.4$, $P(\text{IsBadBuy} = \text{No}) = 0.6$
- Conditional Probability: $P(A_i | C)$
 - If A_i is a categorical variable
 - $P(\text{Auction}=\text{OTHER} | \text{IsBadBuy}=\text{Yes})$
 - A_i is a numeric variable?—Probability density estimation

Proportion of cars with
Auction=OTHER that have
IsBadBuy= Yes

Naïve Bayes

WheelType	Auction	VehicleAge	IsBadBuy
Alloy	OTHER	3	Yes
Special	ADESA	5	No
Alloy	MANHEIM	4	No
unkwnWheel	OTHER	3	No
unkwnWheel	OTHER	6	Yes

Prediction for (WheelType=unkwnWheel, Auction=OTHER, VehicleAge=5)

- $P(\text{IsBadBuy} = \text{Yes} \mid \text{WheelType} = \text{unkwnWheel}, \text{Auction} = \text{OTHER}, \text{VehicleAge} = 5)$
 $\propto P(\text{WheelType} = \text{unkwnWheel} \mid \text{IsBadBuy} = \text{Yes}) * P(\text{Auction} = \text{OTHER} \mid \text{IsBadBuy} = \text{Yes}) * P(\text{VehicleAge} = 5 \mid \text{IsBadBuy} = \text{Yes}) * P(\text{IsBadBuy} = \text{Yes})$
 $= 0.5 * 1 * 0.228 * 0.4 = 0.0456$
- $P(\text{IsBadBuy} = \text{No} \mid \text{WheelType} = \text{unkwnWheel}, \text{Auction} = \text{OTHER}, \text{VehicleAge} = 5)$
 $\propto P(\text{WheelType} = \text{unkwnWheel} \mid \text{IsBadBuy} = \text{No}) * P(\text{Auction} = \text{OTHER} \mid \text{IsBadBuy} = \text{No}) * P(\text{VehicleAge} = 5 \mid \text{IsBadBuy} = \text{No}) * P(\text{IsBadBuy} = \text{No})$
 $= 0.333 * 0.333 * 0.205 * 0.6 = 0.0136$

■ IsBadBuy = Yes (40%; 2 instances)

WheelType:	Alloy	1
	Special	0
	unkwnWheel	1
Auction:	ADESA	0
	MANHEIM	0
	OTHER	2

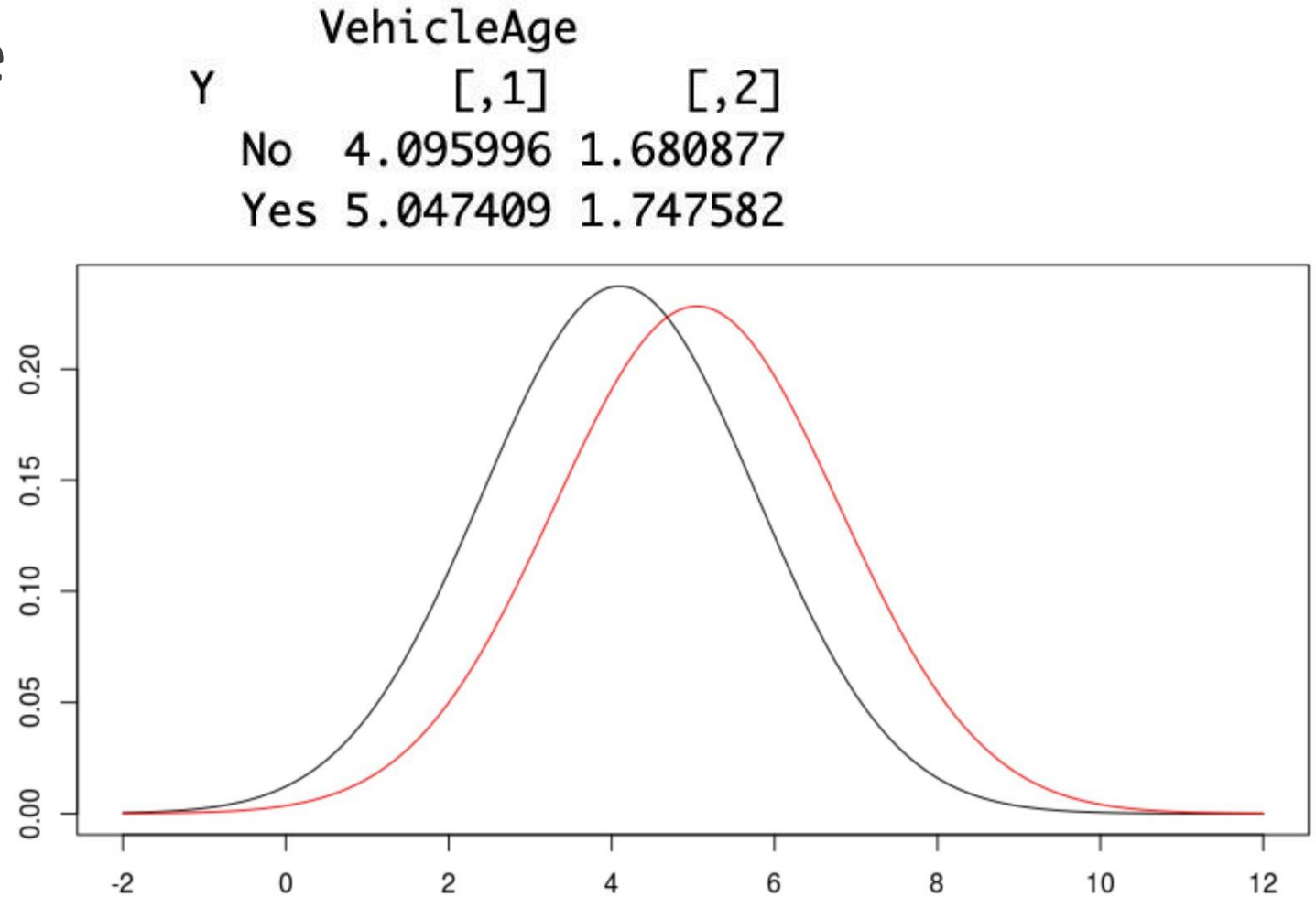
■ IsBadBuy = No (60%; 3 instances)

WheelType:	Alloy	1
	Special	1
	unkwnWheel	1
Auction:	ADESA	1
	MANHEIM	1
	OTHER	1

	VehicleAge	
Y	[,1]	[,2]
No	4.095996	1.680877
Yes	5.047409	1.747582

Naïve Bayes

- For variable VehicleAge
 - If IsBadBuy = No,
 - Mean 4.095996, standard deviation 1.680877
 - If IsBadBuy = Yes,
 - Mean 5.047409, standard deviation 1.747582



Standard Normal Distribution & Z-Scores

- **Standard Normal Distribution**
- A normal distribution with:
 - Mean (μ) = 0
 - Standard deviation (σ) = 1
- Symmetrical, bell-shaped curve
- Total area under the curve = **1** (represents probability)
- **Z-Score:** A **z-score** tells us how many standard deviations a value is from the mean:

$$Z = \frac{x - \mu}{\sigma}$$

- Positive $z \rightarrow$ value is **above** the mean
- Negative $z \rightarrow$ value is **below** the mean

Naive Bayes, Z-Scores, and Numeric Variables

Gaussian Naive Bayes

For numeric features, Naive Bayes assumes values follow a **normal distribution within each class**

For each class C_k , estimate:

- mean μ_k
- standard deviation σ_k

From value to Z-score

$$Z = \frac{x - \mu_k}{\sigma_k}$$

Naive Bayes, Z-Scores, and Numeric Variables

Z measures how far the observed value is from the class mean
Equivalent to standardizing to the **standard normal distribution**

Probability calculation

- Naive Bayes uses the **normal PDF**:

$$p(x \mid C_k) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x - \mu_k)^2}{2\sigma_k^2}\right)$$

This gives the **likelihood** of observing x under class C_k

Naive Bayes, Z-Scores, and Numeric Variables

- **Why PDF, not area (Z-table)?**
- For continuous variables, $P(X = x) = 0$;only **densities** are meaningful
- Z-tables give **cumulative probabilities** $P(X \leq x)$,not point likelihoods
- Classification needs **relative plausibility at x** across classes, which the PDF provides

Naive Bayes Models (scikit-learn)

- Continuous numeric features; assumes a normal distribution per class → *GaussianNB*
- Non-negative count or frequency features (e.g., word counts, TF-IDF) → *MultinomialNB*
- Binary features (0/1), models presence vs absence → *BernoulliNB*
- Categorical (nominal) features with finite categories → *CategoricalNB*
- Text data variant for imbalanced classes; improves MultinomialNB performance → *ComplementNB*
- **Note**
 - Each model uses **one likelihood type for all features**
 - No built-in hybrid Gaussian + Multinomial NB in scikit-learn

Build a Naïve Bayes Model on Training Data

Training data

	IsBadBuy	MMRCurrentAuctionAveragePrice	VehBCost	VehicleAge	...
0	No	2871	5300	8	
1	Yes	1840	3600	8	
2	No	8931	7500	4	
3	No	8320	8500	5	
4	No	11520	10100	5	
...	...	...	...	...	...
6995	No	2762	4185	4	
6996	Yes	5144	8705	5	
6997	No	2912	4485	6	
6998	No	4965	6495	4	
6999	No	2724	3985	6	

Naïve Bayes Model

Generate
count for
each
categorical
variable



Generate
distribution
for each
numeric
variable



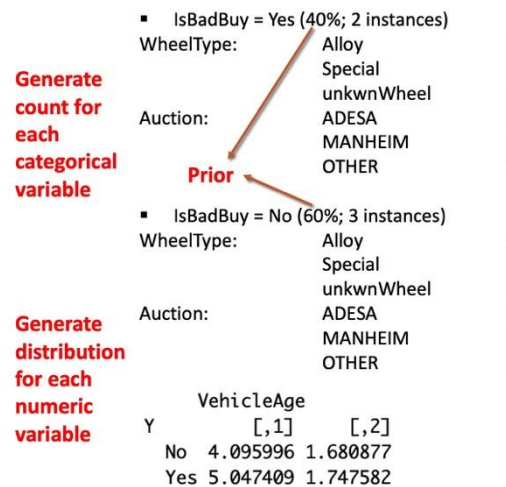
■ IsBadBuy = Yes (40%; 2 instances)
WheelType: Alloy 1
Special 0
unkwnWheel 1
Auction: ADESA 0
MANHEIM 0
OTHER 2

Prior

■ IsBadBuy = No (60%; 3 instances)
WheelType: Alloy 1
Special 1
unkwnWheel 1
Auction: ADESA 1
MANHEIM 1
OTHER 1

VehicleAge
Y [,1] [,2]
No 4.095996 1.680877
Yes 5.047409 1.747582

Evaluate Naïve Bayes Model Performance



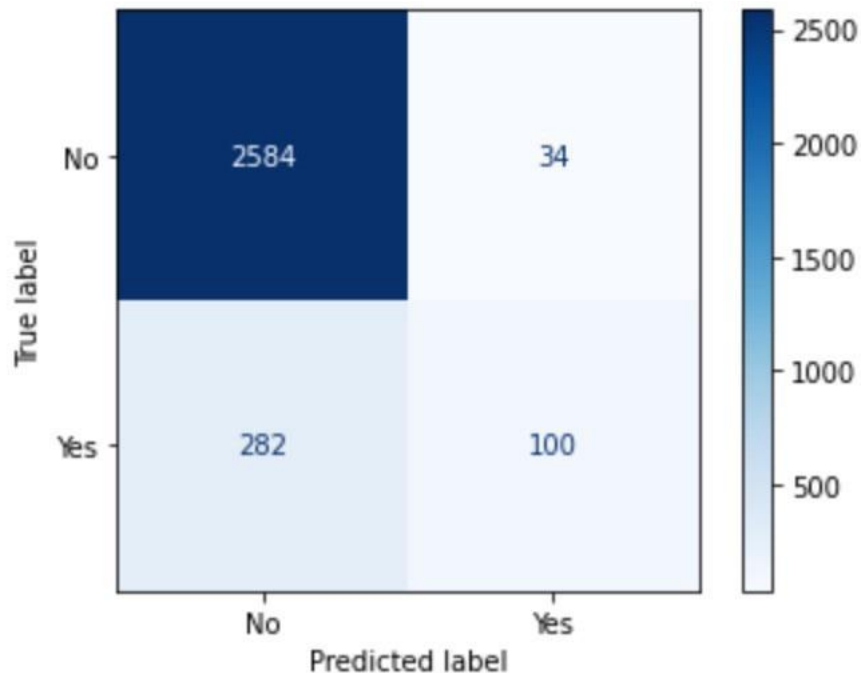
...	...	...	...	...	...	...	...	...	...	...
9995		7536	6600	4	85377	983	0	0	0	
9996		4921	7000	7	89665	1543	0	0	1	
9997		9263	9000	4	59383	1417	0	0	1	
9998		3240	5500	4	48642	482	0	0	0	
9999		6148	7000	5	82563	1243	0	0	0	

Testing data

Predictions	Real value
IsBadBuy	IsBadBuy
No	No
No	No
No	No
No	No
Yes	No
Yes	No

Evaluate Naïve Bayes Model Performance

Performance on the **testing data**:



	precision	recall	f1-score	support
No	0.90	0.99	0.94	2618
Yes	0.75	0.26	0.39	382
accuracy			0.89	3000
macro avg	0.82	0.62	0.66	3000
weighted avg	0.88	0.89	0.87	3000

1. How many cars are predicted as Not bad buy?
2. How many bad buy cars are predicted as Not bad buy?
3. How many bad buy cars are identified correctly by the naive bayes model?
4. If the naive bayes predicts a car as bad buy, what is the probability that such prediction is correct?
5. Does the naive bayes model have better performance on majority (IsBadBuy = 'No') or minority class (IsBadBuy = 'Yes')?

Model Comparison

- Model a classification rule directly
 - Decision Tree: rule-based model
- Make a probabilistic model of data within each class
 - Naïve Bayes: probabilistic model

Model Comparison

- Comparing Decision Tree and Naïve Bayes
 - Decision Tree with max depth = 2

	precision	recall	f1-score	support
No	0.90	1.00	0.94	2618
Yes	0.90	0.22	0.36	382
accuracy			0.90	3000
macro avg	0.90	0.61	0.65	3000
weighted avg	0.90	0.90	0.87	3000

Overall performance
on NOT bad buy class

Overall performance
on bad buy class

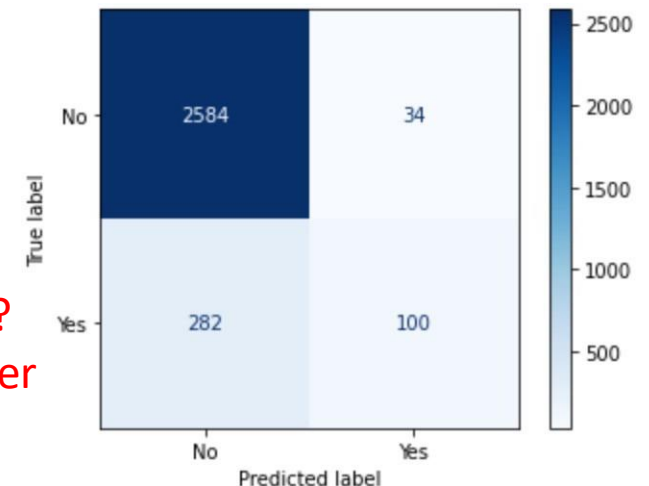
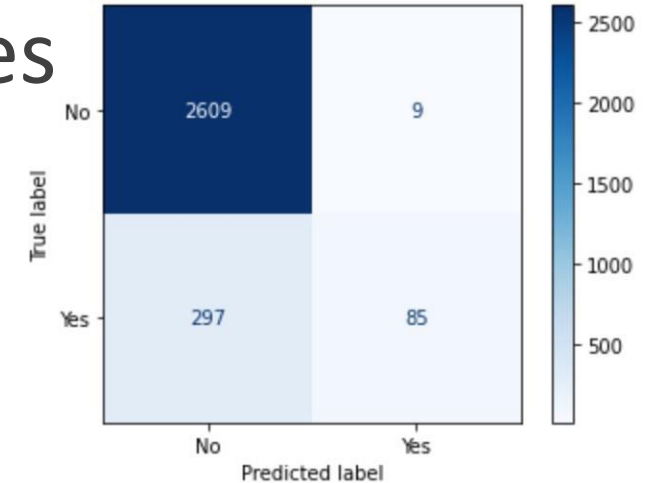
Model's overall
performance

- Naïve Bayes with Laplace smooth = 1

	precision	recall	f1-score	support
No	0.90	0.99	0.94	2618
Yes	0.75	0.26	0.39	382
accuracy			0.89	3000
macro avg	0.82	0.62	0.66	3000
weighted avg	0.88	0.89	0.87	3000

1. Which model has
better performance?

2. Which model is better
to identify bad cars?



Model Comparison

- Overall model performance comparison
 - Accuracy
- Compare performances on each class
 - F-measure: single metrics combines precision and recall and measures the overall performance on each class
 - Precision: confidence/effectiveness of predictions
 - Recall: ability of identifying instances belonging to a class

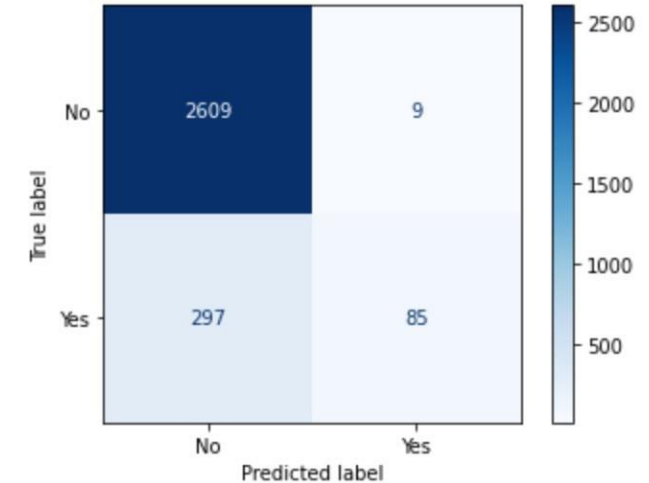
Cost-benefit analysis

Assume the following costs / benefits:

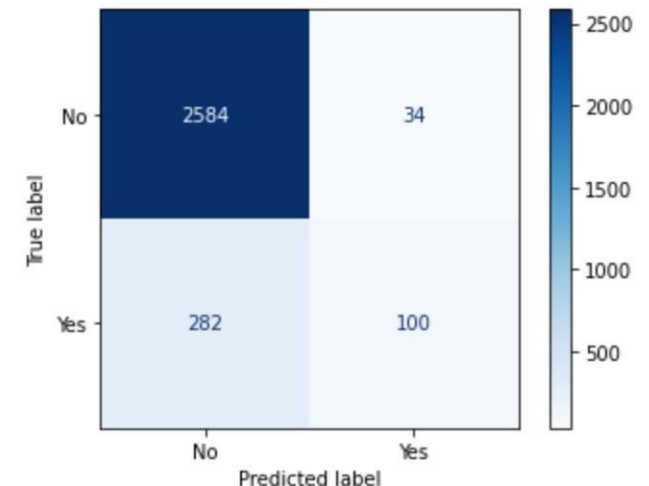
- Cost of buying a bad car: \$2000
- Profit of purchasing a good car: \$500
- Opportunity cost of a good car that is missed: \$500

What is the total net benefit/cost of two models?

Decision Tree with
max depth = 2



Naïve Bayes with
Laplace smooth = 1



Cost-benefit analysis

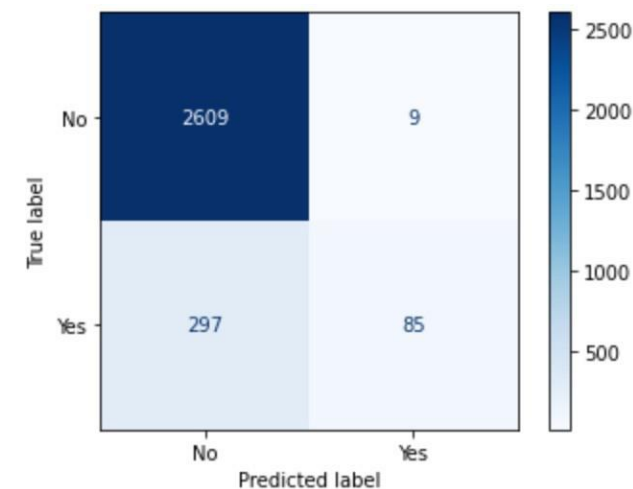
Assume the following costs / benefits:

- Cost of buying a bad car: \$2000
- Profit of purchasing a good car: \$300
- Opportunity cost of a good car that is missed: \$300

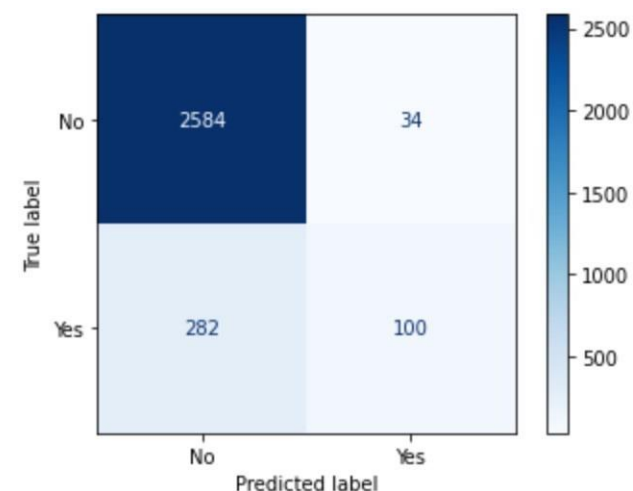
Decision Tree model:

- Cost: $297 * 2000$
- Profit: $2609 * 300$
- Opportunity cost: $9 * 300$
- Profit – Cost - Opportunity cost: 186,000

Decision Tree with
max depth = 2



Naïve Bayes with
Laplace smooth = 1



Cost-benefit analysis

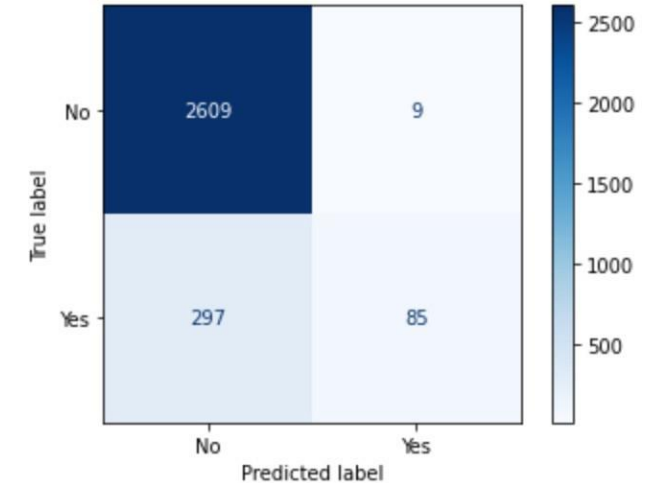
Assume the following costs / benefits:

- Cost of buying a bad car: \$2000
- Profit of purchasing a good car: \$300
- Opportunity cost of a good car that is missed: \$300

Naïve Bayes model:

- Cost: $282 * 2000$
- Profit: $2584 * 300$
- Opportunity cost: $34 * 300$
- Profit – Cost - Opportunity cost: 201,000

Decision Tree with
max depth = 2



Naïve Bayes with
Laplace smooth = 1

